



Original Research

Effect of chest wall mobilization on respiratory muscle function in patients with severe chronic obstructive pulmonary disease (COPD): A randomized controlled trial

Amy Y.Y. Tsui^{a,*}, Rosanna M.W. Chau^a, Gladys L.Y. Cheing^b, Thomas Y.W. Mok^c, S.O. Ling^c, Candy H.Y. Kwan^c, Sharon M.H. Tsang^b

^a Physiotherapy Department, Kowloon Hospital, Hong Kong

^b Department of Rehabilitation Sciences, The Hong Kong Polytechnic University, Hong Kong

^c Department of Respiratory Medicine, Kowloon Hospital, Hong Kong

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ABSTRACT

Background: Clinical trials have demonstrated positive correlation between pulmonary function and chest wall expansion in COPD. Decrease in chest wall expansion in patients with COPD compromises rib cage mobility and functional length of respiratory muscles that ultimately jeopardize the efficacy and function of respiratory system.

Method: Thirty male adults (mean age: 74.97 ± 6.29) suffered with severe COPD were randomly allocated to either experimental group (chest wall mobilizations) or control group. Both groups received standardized education and walking exercise (twice/week) for 6 weeks. Patients in experimental group received additional chest wall mobilizations that include stretching and joints mobilization. Pulmonary function, respiratory muscle strength, thoracic excursion, cervical and thoracic range of movement were evaluated at baseline, post-program and at 3-month follow-up.

Results: There were significantly greater improvements in respiratory muscle strength, thoracic excursion and thoracic range of movement ($p < 0.01$) except thoracic flexion. Lower thoracic excursion is strongly associated with increase in maximum inspiratory pressure ($\beta = 13.64$, $p < 0.001$) and maximum expiratory pressure ($\beta = 16.23$, $p < 0.001$). Thoracic range of movement especially extension ($p < 0.001$) and bilateral rotation ($p < 0.01$) exhibit a strong relationship with increase in lower thoracic excursion (adjusted $R^2 = 0.876$) as shown in multiple regression analysis.

Conclusion: Additional chest wall mobilization in the rehabilitation of patients with COPD is likely to enhance thoracic extension and rotation which increase lower thoracic excursion. This significant improvement in chest expansion capacity allows respiratory muscles to work at an optimal functional length which result in greater respiratory muscle strength in patients with severe COPD.

1. Introduction

One of the hallmarks of COPD is an accelerated decline in lung function, as measured by spirometry [1]. This decline in lung function is mainly related to aging and pathological changes caused by disease processes [2]. Patients with COPD suffered from expiratory airflow limitation, which contributes to emphysematous changes and hyperinflation of the lungs [3]. Lung hyperinflation is a classic manifestation of COPD, and it negatively affects the mechanics of the chest wall, i.e., it

compromises the function of respiratory muscles and ventilatory efficiency [4]. Lung hyperinflation with the chest wall maintained in hyperinflated state leads to an overall decrease in chest wall mobility, which becomes rigid as the disease progresses to severe and very severe COPD [5,6].

Respiratory muscle strength and lung volume are related to chest wall mobility in healthy individuals [7,8]. Earlier studies reported that individuals with chronic diseases such as rheumatoid arthritis and ankylosing spondylitis who have significant reduction in rib cage or

* Corresponding author.

E-mail address: ctyyz01@ha.org.hk (A.Y.Y. Tsui).

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spinal mobility, were found to have substantial decline in lung volume and respiratory muscles strength [9,10]. Meanwhile, reduced chest wall mobility has been found to be more prevalent in severe and very severe COPD [5] and was associated with a decrease in forced expiratory volume in 1 s (FEV₁), forced vital capacity (FVC) and FEV₁/FVC [6]. A cross-sectional study by Heneghan et al. provided evidence that substantiates the significant and strong association between the reduced thoracic and cervical mobility that coexisted with the altered cervico-thoracic posture and poor pulmonary function (FEV₁) in patients with mild to severe COPD [11].

Chest wall mobilization include stretching of the respiratory and chest wall muscles and thoracic joint mobilization, on chest wall mobility and lung function have been investigated for years [12–15]. In de Sá et al. study, patients with severe COPD showed an increase in tidal volume and decrease in end expiratory pressure after a single session of stretching of respiratory muscles [12]. Yelvar et al. demonstrated an immediate improvement in FEV₁, FVC, VC, maximum inspiratory pressure (MIP), and maximum expiratory pressure (MEP) after a single session of cervical, thoracic, sternoclavicular, and scapulothoracic joint mobilization and myofascial release of the respiratory and accessory muscles in patients with severe grade of COPD [15]. Engel et al. demonstrated positive short-term effect of a 4-week manual therapy, including soft tissue therapy and thoracic joint mobilization, on FEV₁ and FVC, in patients with moderate to severe COPD [14]. The RCT pilot trial by Engel et al. demonstrated medium-term effect of additional manual therapy in Pulmonary Rehabilitation Program [13].

While there is evidence to support chest wall mobilization in augmenting the pulmonary function of COPD patients, the sustainability of the effect of chest wall mobilization on pulmonary function in the rehabilitation of COPD would be of interest. Existing literature indicates that the area of musculoskeletal dysfunction is rarely targeted in the management and rehabilitation of patients with COPD. To address the aforementioned gap in the literature, the present study aimed to investigate the cumulative and maintenance effect of chest wall mobilization on respiratory muscle function and its relationships with the musculoskeletal performance of patients with severe COPD.

2. Methods

Participants were recruited from Respiratory Outpatient Clinic of Kowloon Hospital. Patients with a diagnosis of COPD were assessed for their eligibility according to the inclusion and exclusion criteria. The patients were included in the study if they were in severe grade of the disease according to the Global Initiative for Chronic Obstructive Lung Disease (GOLD) spirometric classification (forced expiratory volume in 1 s (FEV₁)/forced vital capacity (FVC) < 0.7 and 30 % ≤ FEV₁ < 50 % predicted), were in a stable clinical state with no acute exacerbation require hospitalization from recent 1 month, ambulatory and communicable. Exclusion criteria were comorbidities causing poor mobility or ambulation, current smoker and contraindications to chest wall mobilization (e.g. severe and unstable rib fracture, metastatic bone cancer, tuberculosis spondylitis, and severe osteoporosis). Written informed consents were obtained from all participants prior to the commencement of this study.

A pre-intervention (baseline) assessment was conducted in all patients by a single physiotherapist, who was blinded to the group allocation. Respiratory function (i.e. FEV₁, FVC, MIP and MEP), and musculoskeletal performance (i.e. cervical and thoracic range of movement and thoracic excursion) were assessed. The patients were divided into 2 groups: chest wall mobilization group (CWMG) or control group (CG). Allocation to the CWMG or CG was randomized in a 1:1 ratio and was concealed from all patients, with each patient selecting a sealed, opaque envelope indicating the group. A project assistant who was not aware of the details of the research study was responsible for the randomization and group allocation procedures.

Patients in the CWMG received the studied intervention involving

the structured chest wall mobilization program, which comprised of three therapist-assisted stretching exercises and one thoracic joint mobilization (Fig. 1).

2.1. Chest wall mobilization techniques

Therapist-assisted stretching for chest wall mobilization was performed according to the guidelines for general stretching of skeletal muscles [16]. Stretching was performed bilaterally. The therapist assisted the movement to the possible range, and the patient's arms were held in a position with the muscle in tension to a point of a stretching sensation. The stretching was performed with two sets of ten consecutive incursions in each position and a 1-min interval for rest between the series. The position in which the patient felt the stretching sensation was held for 10 s [17]. The stretching exercises took approximately 30 min.

2.2. Stretching exercise 1: lateral flexion of the chest wall

This technique is shown in Fig. 1 (i). The patient was instructed to lie down in a side-lying position with a foam roller placed in the infra-axillary region. The forearms were flexed, with the hands resting along the occipital region. The therapist used the palmar region of both hands to mobilize the ribs in the cranial and caudal directions [18].

2.3. Stretching exercise 2: chest wall extension

This technique is shown in Fig. 1 (ii). The patient was instructed to lie down in a supine position on a foam roller placed in the mid-thoracic region. The arms were raised over the head, and the patient grasped a fixed wall bar placed approximately 10 inches from the surface of the plinth [19].

2.4. Stretching exercise 3: pectoralis major stretching

This technique is presented in Fig. 1 (iii). The patient was instructed to lie down in a supine position. The arms were abducted at approximately 135° with the elbows extended. The movement was performed with the therapist's hands on the patient's forearm, stretching the upper chest in the direction of the pectoralis major muscle fibers [12].

2.5. Thoracic joint mobilization: lateral gliding of the thoracic spine

This technique is presented in Fig. 1 (iv). Thoracic joint mobilization involved non-thrust transverse vertebral pressure, as described by Maitland and his co-workers [20]. The patient was instructed to lie down in the prone position with the arms on the side and the head in a "forehead rest position." Mobilization was applied to the whole thoracic spine. The therapist stood at the level of the thoracic vertebra to be mobilized on one side of the patient. The pad of the therapist's non-dominant thumb was placed in contact with the lateral aspect of the spinous process of T1, whereas the dominant thumb was placed on the dorsal side of the other thumb. Pressure was applied to the spinous process to produce small amplitude, low velocity oscillations in resistance to the end range of the vertebra (Grades IV). This procedure was performed for 30 s, following which it was sequentially applied to the next caudal level. The same pattern of application was used on the patient's contralateral side.

Patients in the CG were required to lie down in the supine, side lying, and prone positions for approximately 30 min so that the treatment time and effect of positioning were standardized across both groups. In addition, both the CWMG and CG received standardized educational classes (a total of 3 sessions) and 15 min of treadmill walking exercise at a low speed with SpO₂ ≥ 88 % in each session.

It was a 6-week program with two sessions/week (i.e., a total of 12 sessions) in accordance with the American College of Sports Medicine's [17] recommendation for exercise training programs in patients with

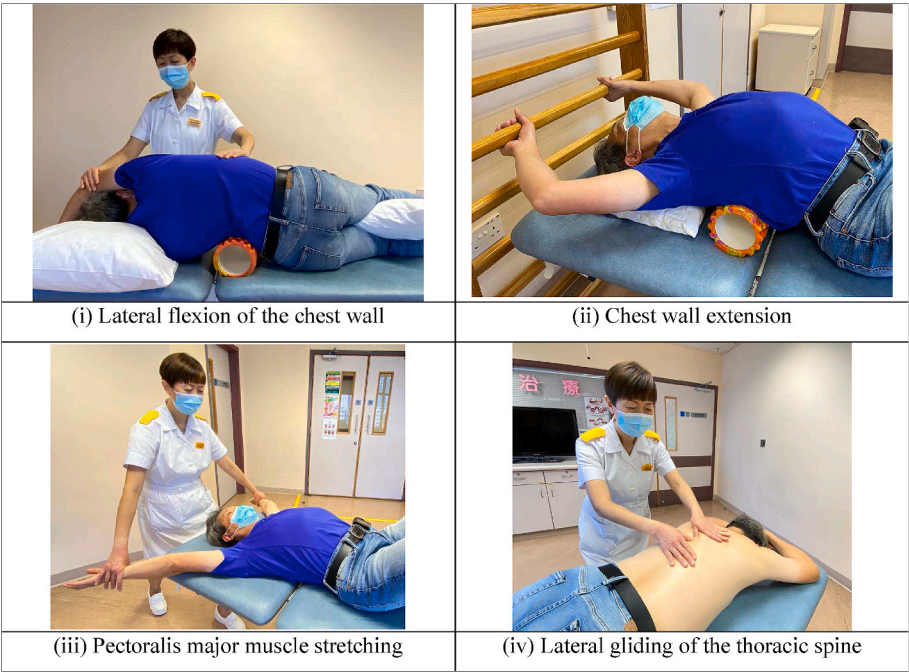


Fig. 1. Chest wall mobilization techniques.

COPD. A post-intervention evaluation was conducted upon program completion in the sixth week. A follow-up session was conducted at 3 months post-intervention to evaluate the maintenance effects of chest wall mobilization on respiratory function and musculoskeletal performance. The procedures were conducted according to the Consolidated Standards of Reporting Trials (CONSORT) guidelines and are summarized in Fig. 3.

In addition, for the maintenance effects of intervention and for further improving chest wall mobility as well as soft tissue elasticity, self-stretching exercises involving trunk extension, rotation, and side flexion, as well as stretching of the pectoralis major muscle, were taught to the patients as part of a home exercise program (Fig. 2). Patients were

asked to repeat each exercise for 10 times, hold the stretch for 10 s at end range of movement, then, have a 10 s break and move to the next exercise. Patients were instructed to do the exercises 3 times/week for 6 weeks. The home stretching exercise program of CG was limited to upper and lower limb stretching. An exercise logbook with diagrams of the stretching exercises was provided to the patients for them to follow the exercises and recorded their participation. The patients were contacted weekly by telephone in order to motivate and encourage them to adhere to the program and fill in the logbook.

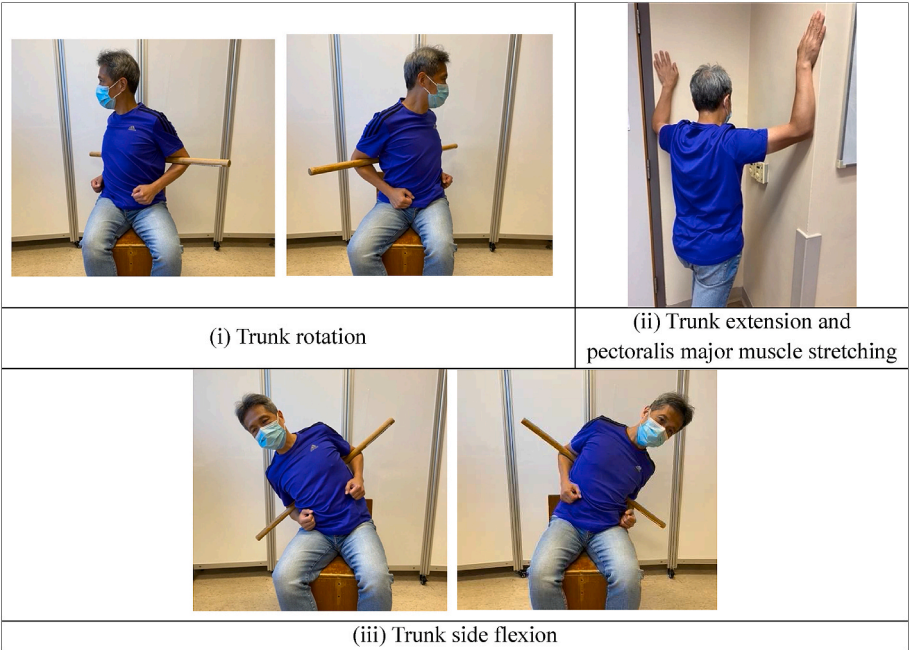


Fig. 2. Home exercise program for the chest wall mobilization group.

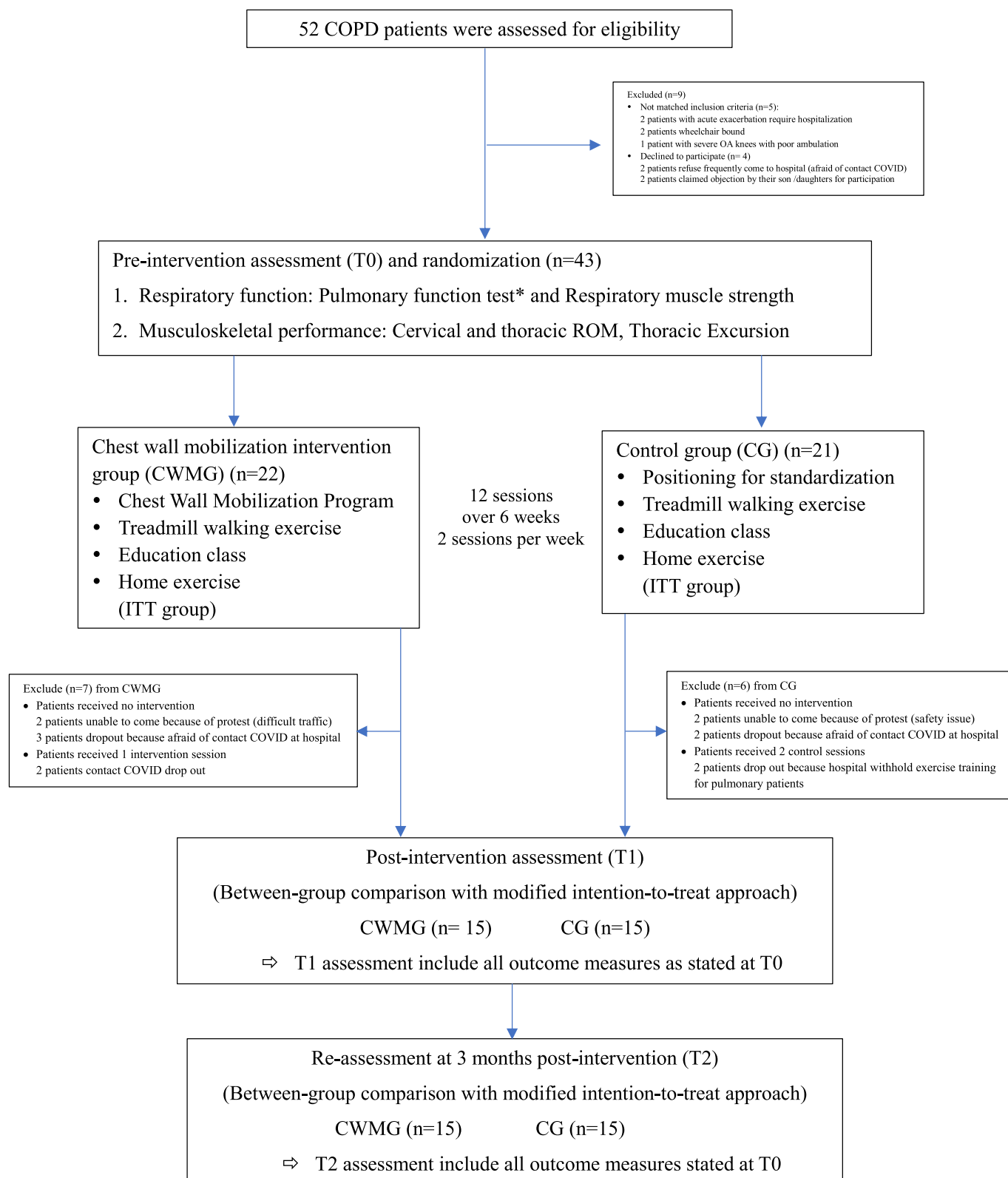


Fig. 3. CONSORT diagram of the study

Abbreviations: COPD, chronic obstructive pulmonary disease; ROM, range of movement;

*Forced Vital Capacity (FVC) is the primary outcome of the study.

2.6. Outcome measures

2.6.1. Pulmonary function test

The pulmonary function test was performed using a portable

spirometer (Spirolab III; Medical International Research, Italy) according to the American Thoracic Society guidelines [21]. The value of FEV₁ and FVC were obtained from FVC maneuver. Three measurements were taken with 30s rest between trials. The best values were used for

analysis.

2.6.2. Assessment of respiratory muscle strength

Maximal inspiratory pressure (MIP) and maximal expiratory pressure (MEP) were assessed by using an electronic pressure transducer (MicroRPM; Micro Medical Ltd., Kent, UK). MIP was measured at residual volume, and MEP was measured from total lung capacity according to Respiratory Pressure Meter Operating Manual [22]. While previous studies only affirmed that Micro RPM is a reliable and valid instrument for assessing MIP & MEP and with good test/retest reliability of maximum mouth pressure measurements performed by the healthy individuals [23,24], an excellent intra-rater reliability in MIP and MEP measurements was established in this study ($n = 5$, $ICC = 0.85-0.89$). The measurement of MIP and MEP were repeated three times with at least 30s rest after each test and the best values were used for analysis.

2.6.3. Cervical and thoracic movement measured using a three-dimensional motion sensor

The regional mobility of the cervical and thoracic spine was measured using an inertial measurement system, the MyoMotion system (Noraxon Inc., USA) at a sampling frequency of 1500 Hz. Three motion sensors were placed over the external occipital protuberance and spinous processes of first thoracic vertebra (T1) and 12th thoracic vertebra (T12), respectively, for measuring the movement of the cervical and thoracic spine region. Previous studies have reported good to excellent reliability in measuring regional spine mobility using the same protocol [25,26]. In this study, similar intra-rater reliability was obtained in measuring cervical ($n = 5$, $ICC = 0.85-0.91$) and thoracic ROM ($n = 5$, $ICC = 0.88-0.93$).

2.6.4. Thoracic excursion

Thoracic excursion is usually called chest expansion measurement in the literature and has been demonstrated to be well correlated with lung function parameters (FEV_1/FVC [$r = 0.68$]) [6] and respiratory muscle strength (MIP & MEP [$r = 0.32$]) [8]. Upper chest expansion measurement was found to be useful for the evaluation of chest mobility in clinical practice and provides indirect information on lung volume and inspiratory muscle strength [27]. In this study, the intra-rater reliability in UpTxExc ($n = 5$, $ICC = 0.89-0.92$) and LoTxExc ($n = 5$, $ICC = 0.80-0.82$) measurements were high. The mean measurement of upper and lower thoracic excursions of the 5 subjects was 2.30 cm and 3.10 cm and SEM is 0.5 cm and 0.6 cm respectively of three trials.

A measuring tape was used to measure thoracic excursion in centimeters (cm) at two levels of the rib cage. For measuring upper thoracic excursion, the anatomical landmarks used were the spinous process of T5, midpoint of the clavicular line, and third intercostal space [28]. For measuring lower thoracic excursion, the anatomical landmarks used were the spinous process of T10 and xiphoid process [28]. To calculate thoracic excursion, the inspiratory diameter was subtracted from the expiratory diameter. Three trials of each thoracic excursion measurement were performed, and the average value was used in the analysis.

2.7. Ethical consideration

Ethical approval was obtained from The Hong Kong Polytechnic University and Research Ethics Committee (Kowloon Central Cluster/Kowloon East) of the Hong Kong Hospital Authority (reference no: KC/KE-18-0144/FR-3). This study was registered at [ClinicalTrials.gov](https://www.clinicaltrials.gov) (ID: NCT04122365).

2.8. Statistical analysis

Statistical analyses were performed using Statistical package for Social Science (SPSS), Version 23.0 (IBM SPSS Inc.). Independent *t*-test was used to compare the between group differences of demographic data. Two-way repeated measures analysis of variance (ANOVA) was

used to detect changes in all the outcome measures at various timepoints in comparison to the baseline measurements. The interaction and main effects between the two study groups across all timepoints were analyzed for all the outcome measures. A post hoc test was used for individual pairs of group comparisons if a significant ANOVA was detected, and the Bonferroni correction was used to adjust the inflation of alpha caused by multiple comparisons. In addition, correlation analysis was used to analyze the direction and strength of relationships among outcome variables related to thoracic mobility, thoracic excursion and respiratory muscle functions. Multiple linear regression with the backward approach was used to explore the interactions among different outcome variables.

3. Results

3.1. Demographics and characteristics of the patients

A total of 30 male patients completed the program and assessments. All the patients had severe COPD (GOLD stage III), as determined by spirometry ($30\% \leq FEV_1 < 50\%$ predicted) [3]. There were no significant differences in demographic characteristics and medication between the two groups. The mean age of the patients was 74.97 ± 6.29 years with height 1.64 ± 0.41 m and weight 56.78 ± 8.26 kg. They were evenly distributed in B, C and D group in the ABCD assessment with similar mMRC (Modified Medical Research Council) Dyspnea Scale 1.87 ± 0.7 . The two groups of patients were receiving similar bronchodilators and anti-inflammatory agents such as β -2 Agonists, LABA, LAMA, ICS or antimuscarinic agents.

3.2. Baseline characteristics of the patients

Baseline measurement (mean $\pm$ SD) of the patients are presented in Table 1. No significant differences ($p > 0.05$) were detected on the outcome measures at baseline between the CWMG and CG.

Table 1
Baseline characteristics of the patients in CWMG and CG.

Baseline Measurement	CWMG (n = 15)	CG (n = 15)	p-value
Pulmonary Function			
FEV ₁	0.84 ± 0.14	0.92 ± 0.18	0.178
FVC	1.65 ± 0.04	1.64 ± 0.44	0.885
Respiratory Muscle Strength			
MIP	50.67 ± 14.77	50.60 ± 14.86	0.990
MEP	80.53 ± 9.65	80.07 ± 9.67	0.896
Cervical ROM			
Flexion	44.97 ± 5.53	45.09 ± 5.66	0.956
Extension	32.69 ± 10.28	34.29 ± 9.51	0.662
(L) Rotation	58.61 ± 5.56	58.57 ± 5.61	0.984
(R) Rotation	58.38 ± 5.17	58.22 ± 5.39	0.932
(L) Side Flexion	26.99 ± 5.90	28.89 ± 5.03	0.351
(R) Side Flexion	26.61 ± 6.11	28.63 ± 4.97	0.327
Thoracic ROM			
Flexion	23.34 ± 2.58	26.17 ± 5.58	0.086
Extension	14.11 ± 2.33	14.27 ± 2.26	0.850
(L) Rotation	35.95 ± 2.99	37.15 ± 6.62	0.530
(R) Rotation	36.03 ± 3.25	37.39 ± 6.54	0.479
(L) Side Flexion	21.43 ± 5.27	21.47 ± 5.29	0.984
(R) Side Flexion	21.38 ± 5.61	21.67 ± 5.23	0.886
Thoracic Excursion			
Upper	2.31 ± 0.80	2.47 ± 0.83	0.611
Lower	3.00 ± 0.97	2.95 ± 0.94	0.879

The p-values represent the results of the comparisons between the two groups. An independent *t*-test was used for the between-group comparisons. The level of significance was set at $p = 0.05$.

Abbreviations: CWMG, chest wall mobilization group; CG, control group; FEV₁, forced expiratory volume in the first second; FVC, forced vital capacity; MIP, maximum inspiratory pressure; MEP, maximum expiratory pressure; ROM, range of movement.

3.3. Respiratory function

Respiratory function included assessment of pulmonary function and respiratory muscle strength.

3.3.1. Pulmonary function test

When comparing the pulmonary function results obtained between CWMG and CG across the 3 timepoints, there was no significant interaction between time and group in FEV₁ and FVC ($p > 0.05$).

3.3.2. Respiratory muscle strength

In the CWMG, MIP and MEP improved significantly across time as assessed at the post-intervention and 3-month follow-up assessment ($p < 0.001$). In the CG, there was no significant difference in MIP and MEP across the three assessment timepoints ($p > 0.05$). There was significant time and group interaction in the case of both MIP and MEP ($p < 0.001$) with significant between-group differences observed at T1 (MIP: $p = 0.002$, MEP: $p < 0.001$) and T2 (MIP: $p = 0.007$, MEP $p < 0.001$).

3.4. Musculoskeletal performance

Musculoskeletal performance included assessment of regional spinal mobility and thoracic excursion.

3.4.1. Cervical ROM

When comparing the cervical ROM between the CWMG and CG across the 3 timepoints, the between-group differences in all the cervical ROM measurements were non-significant ($p > 0.05$).

3.4.2. Thoracic ROM

There was significant time-and-group interaction in the case of all the thoracic mobility measurements ($p < 0.001$) except for thoracic flexion. In CWMG, thoracic extension, bilateral thoracic rotation and thoracic side flexion increased significantly across time as assessed at the post-intervention and 3-month follow-up assessment ($p < 0.001$). The most marked post-intervention improvement was seen in bilateral thoracic side flexion and thoracic extension in which there was an overall increase of 29.7 % and 25.4 %, respectively. Further, there was a 9.7 % and 9.4 % improvement in left and right thoracic rotation, respectively. By contrast, in the CG, there was no significant difference in thoracic range of movement across the three assessment timepoints ($p > 0.05$).

3.4.3. Thoracic excursions

In the CWMG, there was significant time-and group interaction of both upper and lower thoracic excursions (UpTxExc and LoTxExc) ($p < 0.01$). There was significant between-group difference in LoTxExc at T1 ($p = 0.002$) and T2 ($p = 0.005$). However, there was non-significant between-group difference in UpTxExc at T1 and T2 ($p > 0.05$). In the CWMG, both UpTxExc and LoTxExc increased significantly across time, the UpTxExc and LoTxExc increased by 0.36 cm and 1.0 cm respectively at 3-month follow-up. The mean change of 1.0 cm in LoTxExc exceeded the MCID of 0.8–0.9 cm, suggested in a previous study involving patients with chronic respiratory disease [29]. However, the mean change of 0.36 cm in UpTxExc was far below the suggested MCID. In the CG, there was non-significant within-group difference in both the UpTxExc and LoTxExc ($p > 0.05$).

All the significant results of respiratory muscle strength, thoracic range of movement and thoracic excursion were presented in Table 2.

3.5. Relationships among thoracic mobility, thoracic excursion and respiratory muscle strength

The relationships among the outcome variables related to thoracic mobility, thoracic excursion, and respiratory muscle strength are presented in Table 3. There was a strong positive relationship between an

Table 2

Evaluation of respiratory muscle strength, thoracic range of movement & thoracic excursion.

	Experimental Group	Control Group
Maximum Inspiratory Pressure		
Baseline	50.67 ± 14.78	50.60 ± 14.86
Post-treatment	67.27 ± 12.30	50.20 ± 14.75
3 months follow-up	65.53 ± 12.85	50.73 ± 14.90
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.047^*$; time × group interaction: $p < 0.001^*$		
Maximum Expiratory Pressure		
Baseline	80.53 ± 9.65	80.07 ± 9.67
Post-treatment	97.93 ± 9.11	80.53 ± 10.12
3 months follow-up	98.33 ± 8.86	80.13 ± 9.53
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.001^*$; time × group interaction: $p < 0.001^*$		
Thoracic Flexion		
Baseline	23.34 ± 2.58	26.17 ± 5.58
Post-treatment	23.79 ± 2.87	26.00 ± 5.49
3 months follow-up	23.83 ± 2.47	25.97 ± 5.44
Overall within-group effect: $p = 0.663$; overall between group effect: $p = 0.137$; time × group interaction: $p = 0.124$		
Thoracic Extension		
Baseline	14.11 ± 2.33	14.27 ± 2.26
Post-treatment	17.81 ± 2.29	14.34 ± 2.07
3 months follow-up	17.69 ± 2.09	14.23 ± 2.19
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.008^*$; time × group interaction: $p < 0.001^*$		
Thoracic Left Rotation		
Baseline	35.95 ± 2.99	37.15 ± 6.62
Post-treatment	39.73 ± 3.17	37.67 ± 5.89
3 months follow-up	39.45 ± 3.52	37.25 ± 6.19
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.574$; time × group interaction: $p < 0.001^*$		
Thoracic Right Rotation		
Baseline	36.03 ± 3.25	37.39 ± 6.54
Post-treatment	39.92 ± 3.41	37.17 ± 6.14
3 months follow-up	39.41 ± 3.51	37.39 ± 6.09
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.537$; time × group interaction: $p < 0.001^*$		
Thoracic Left Side flexion		
Baseline	21.43 ± 5.27	21.47 ± 5.29
Post-treatment	27.72 ± 3.57	21.42 ± 5.50
3 months follow-up	27.80 ± 3.35	21.53 ± 5.27
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.022^*$; time × group interaction: $p < 0.001^*$		
Thoracic Right Side flexion		
Baseline	21.38 ± 5.61	21.67 ± 5.23
Post-treatment	27.55 ± 3.72	21.97 ± 5.16
3 months follow-up	27.63 ± 3.65	21.94 ± 5.32
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.044^*$; time × group interaction: $p < 0.001^*$		
Upper Thoracic Excursion		
Baseline	2.31 ± 0.80	2.47 ± 0.83
Post-treatment	2.65 ± 0.64	2.49 ± 0.80
3 months follow-up	2.67 ± 0.65	2.43 ± 0.81
Overall within-group effect: $p = 0.002^*$; overall between group effect: $p = 0.765$; time × group interaction: $p = 0.001^*$		
Lower Thoracic Excursion		
Baseline	3.00 ± 0.97	2.95 ± 0.94
Post-treatment	4.12 ± 0.88	2.95 ± 0.97
3 months follow-up	4.00 ± 0.91	2.95 ± 0.98
Overall within-group effect: $p < 0.001^*$; overall between group effect: $p = 0.035^*$; time × group interaction: $p < 0.001^*$		

* $p < 0.05$

increase in MIP and MEP and thoracic extension, thoracic rotation and side flexion. Regarding thoracic excursion, an increase in upper and lower thoracic excursion exhibited a strong positive relationship with MIP, MEP and all thoracic ROM except thoracic flexion.

3.6. Influence of thoracic excursion on respiratory muscle strength and their relationship with thoracic ROM

Multiple linear regression with a backward approach was performed

Table 3

Results of the correlation analysis among the outcomes variables related to respiratory muscle strength, thoracic mobility and thoracic excursion.

Pearson's correlation	Diff_MIP	Diff_MEP	Diff_TxF	Diff_TxE	Diff_Tx Rot [#]	Diff_Tx SF [#]	Diff_Up TxExc	Diff_Lo TxExc
Diff_MIP		0.877**	0.394*	0.909**	0.842**	0.902**	0.526**	0.909**
Diff_MEP	0.877**		0.213	0.811**	0.850**	0.801**	0.592**	0.857**
Diff_TxF	0.394*	0.213		0.388*	0.186	0.476**	0.255	0.302
Diff_TxE	0.909**	0.811**	0.388*		0.812**	0.846**	0.463**	0.920**
Diff_TxRot	0.842**	0.850**	0.186	0.812**		0.777**	0.680**	0.860**
Diff_TxSF	0.902**	0.801**	0.476**	0.864**	0.777**		0.578**	0.831**
Diff_UpTxExc	0.526**	0.592**	0.255	0.463**	0.680**	0.578**		0.588**
Diff_LoTxExc	0.857**	0.857**	0.302	0.920**	0.860**	0.831**	0.588**	

*p < 0.05, **p < 0.01.

Abbreviations: Diff, difference between the results at pre-intervention and the 3-month follow-up; MIP, maximum inspiratory pressure; MEP, maximum expiratory pressure; TxF, thoracic flexion; TxE, thoracic extension; TxRot, thoracic rotation; TxSF, thoracic side flexion; UpTxExc, upper thoracic excursion; LoTxExc, lower thoracic excursion.

[#]There is no significant difference in rotation and side flexion between both sides, the average value of both side of the respective direction of mobility is presented.

to assess (i) the influence of upper thoracic excursion (UpTxExc) and lower thoracic excursion (LoTxExc) on MIP and MEP and (ii) the relationship of thoracic ROM with thoracic excursion. The UpTxExc yielded non-significant p-value for both the MIP and MEP. There was a strong positive relationship between the increase in MIP and MEP and increase in LoTxExc (MIP: $F [30,18] = 132.954$, $p < 0.001$, adjusted $R^2 = 0.820$; MEP: $F [30,18] = 77.679$, $p < 0.001$, adjusted $R^2 = 0.726$). Moreover, the increase in LoTxExc was positively and strongly associated with the increase in thoracic extension and rotation ($F [4,31] = 103.055$, $p < 0.001$, adjusted $R^2 = 0.876$) in the linear regression analysis.

4. Discussion

To the best of the author's knowledge, this is the first RCT to investigate the cumulative and maintenance effects of a 6-week chest wall mobilization program on respiratory function and musculoskeletal performance in patients with severe COPD. Previous studies have primarily examined either the immediate [12,15] or short-term [13,14] effects. The present findings have provided the positive and superior cumulative effects of a 6-week chest wall mobilization on thoracic excursion, thoracic mobility and respiratory muscle strength in patients with severe COPD.

The passive stretching of the chest wall and thoracic joint mobilization performed by physiotherapist were well tolerated without any complaints in patients in the CWMG although some patients needed to get accustomed to lying down on the foam roller. No adverse effects, including muscle soreness, pain, or stiffness, were reported during and after the mobilization sessions. The chest wall mobilization techniques used in this study were therefore considered as safe, feasible and effective interventions for patients with severe COPD.

4.1. Chest wall mobilization enhances musculoskeletal performance and respiratory muscle strength

Compared with patients in the CG, those in the CWMG exhibited a significant post-intervention improvement in thoracic excursion, which was especially profound in their lower thoracic region. It is essential to note that the improvement obtained in this RCT was indeed greater than the MCID (1-cm increase in the lower thoracic excursion) reported previously for chronic respiratory disease [29]. Chest expansion was found to be positively associated with the respiratory muscle strength in healthy non-smoker [8], and our results support this finding, as the increase in respiratory muscle strength aligns with an increase in thoracic excursion. Specifically, MIP and MEP were strongly associated with an increase in lower thoracic excursion.

Previous studies have demonstrated that a 3–8 week stretching exercise program increased muscle extensibility and decreased muscular tension in healthy individuals [31,32]. Accordingly, the positive findings in our study suggest that stretching the chest wall muscles increases

soft tissue extensibility and reduces muscular tension in the respiratory muscles, which enhances thoracic expansion. When thoracic excursion increases, and the thoracic cage is in a less hyperinflated state, the intercostal muscles and diaphragm work at a more favorable length, which improves muscle efficiency and decreases the work of breathing [4]. Our 6-week chest wall mobilization program had significant effects in augmenting thoracic excursion, thereby promoting respiratory muscle strength.

Previous study has shown that lung hyperinflation alters thoracic geometry, resulting in a descending flattened diaphragm in patients with severe COPD [33]. A shortened diaphragm leads to a decrease in lower rib cage expansion, and inspiratory expansion becomes severely compromised [34]. In our study, patients in the CWMG exhibited a significant increase in lower thoracic excursion. This improvement allows for better lower rib cage expansion, which in turn improves inspiratory expansion; this may contribute to rectifying the altered position of the diaphragm, leading to enhanced breathing mechanics with an improvement in respiratory muscle strength. In addition, an increase in lower chest expansion allows better contraction of the diaphragm, facilitating more effective diaphragmatic breathing in COPD patients.

The results of the study have clearly demonstrated that stretching of the chest wall together with thoracic joint mobilization results in a significant increase in both thoracic excursion and thoracic ROM, apart from during thoracic flexion. A previous study has shown that a significant reduction in thoracic rotation is associated with poor pulmonary function in patients with COPD [11]; therefore, it was reasonable to believe that an increase in thoracic ROM could promote pulmonary function. Our study revealed a strong positive relationship between an increase in MIP and MEP and an increase in lower thoracic excursion. Moreover, an increase in lower thoracic excursion was positively and strongly associated with an increase in thoracic extension and rotation ($F [4,31] = 103.055$, $p < 0.001$, adjusted $R^2 = 0.876$). Our results concur with those of Heneghan's study on the relationship between thoracic rotation and pulmonary function in terms of respiratory muscle strength. Furthermore, any intervention or technique for improving thoracic extension and rotation can be focused on in order to achieve a more pronounced improvement in lower thoracic excursion, which is strongly associated with an increase in MIP and MEP.

In our study, there was no significant improvement in the lung function parameters, FEV₁ and FVC. As expected, FEV₁ remained similar across the assessment timepoints, as the intervention was not designed to have an effect on airflow limitation. In the case of FVC, previous studies involving spinal manipulation and myofascial release or friction massage have reported significant positive effects in COPD patients. In this study, the chest wall mobilization technique involved chest wall stretching and thoracic joint mobilization but not thoracic manipulation. The difference in interventions may be one of the factors that contributed to the non-significant result in terms of FVC. Besides, our results showed increase in lower thoracic excursion (a mean change of 1

cm) which reflect increase in lung volume during breathing. However, as suggested in previous study [6], there should be a large enough change in thoracic excursion to exhibit a significant lung volume change measured in pulmonary function test [6]. Nevertheless, an increasing trend in FVC was observed in this study across time, but it was not significant when compared between the CWMG and CG. Additional evidence is required to confirm whether the chest wall mobilization techniques involved in this study have an effect in improving FVC. The optimum frequency and duration of chest wall mobilization, the choice of mobilization or manipulation techniques maybe the contributing factors and warrant further investigation.

4.2. Chest wall mobilization and pulmonary rehabilitation

The chest wall mobilization techniques used in this study are simple to execute and do not require costly equipment. The effects are cumulative and long lasting at least for 3 months post-intervention. Self-stretching exercises at home are encouraged to maintain the effectiveness of the intervention. However, these techniques are not frequently used in the rehabilitation of COPD patients. Some may argue that these techniques require too much of the therapist's attention, as all the techniques are therapist-assisted. Time constraints may be one of the reasons why these techniques are not popular.

The content of hospital-based pulmonary rehabilitation program for chronic respiratory conditions seldom involves chest wall mobilization exercises. It is difficult to further enhance the effectiveness of cardio-pulmonary training in a pulmonary rehabilitation program if the lungs cannot expand adequately in a restricted and tight chest wall, as is the case in severe and very severe COPD. The present findings advocate the inclusion of chest wall stretching exercises and thoracic joints mobilization to the rehabilitation programme for COPD for maintaining both the chest wall mobility and chest expansion capacity. In this study, the chest wall mobilization techniques were demonstrated to be effective in improving respiratory muscle strength, thoracic excursion and thoracic range of movement in patients with severe COPD. This contribution would be encouraging in the management of chronic respiratory diseases.

4.3. Significance of the study

A strong positive association was demonstrated between lower thoracic excursion and thoracic extension and rotation. Exercises and mobilization or manipulation techniques, which can increase thoracic extension and rotation, should be promoted in patients with COPD in order to increase chest expansion and in turn improve respiratory muscle strength.

4.4. Limitation of the study and future direction

The participants in this study were only elderly male patients with COPD; these patients were selected to avoid the effects of sex and age on the outcome measurements. Therefore, the results of our study cannot be generalized to elderly female or younger male and female patients with COPD.

In this study, the post-treatment effect was observed for 3 months in the severe group of COPD patients. Future studies to involve a wide range of patients such as moderate to severe COPD for a longer follow-up period for capturing re-admission rate and hospitalization days on acute exacerbation of COPD is valuable to observe the long-term effect of chest wall mobility on COPD patients. As hospitalization and frequent re-admission poses great burden to the medical system, affirmative results provide directional information to enhance non-pharmacological management of COPD.

5. Conclusion

Enhancement of thoracic extension and rotation increases lower thoracic excursion after chest wall mobilizations. The significant improvement in chest expansion capacity allows respiratory muscles to work at a more optimal functional length which result in greater respiratory muscle strength in patients with COPD. Thus, it would be beneficial to have chest wall mobilization to be included as a core component in respiratory rehabilitation to optimize the rehabilitation outcomes for patients with severe COPD.

CRediT authorship contribution statement

Amy Y.Y. Tsui: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration, Funding acquisition, Acquisition. **Rosanna M.W. Chau:** Supervision, Resources, Funding acquisition. **Gladys L.Y. Cheing:** Conceptualization, Methodology, Validation, Formal analysis, Supervision, Resources, Writing – review & editing. **Thomas Y.W. Mok:** Resources, Project administration, Visualization. **S.O. Ling:** Resources, Project administration, Visualization. **Candy H.Y. Kwan:** Resources, Project administration, Visualization. **Sharon M.H. Tsang:** Conceptualization, Methodology, Validation, Formal analysis, Supervision, Resources, Writing – review & editing.

Declaration of competing interest

The authors report no declarations of interest. This research had received the Kowloon Central Cluster Research Grant from The Hong Kong Hospital Authority.

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